

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	3 October 2026
Team ID	SWTID-2026-3698
Project Name	Auto Ticket Classification using Flow Designer
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Solution: Auto ticket classification for a school IT helpdesk using ServiceNow Flow Designer

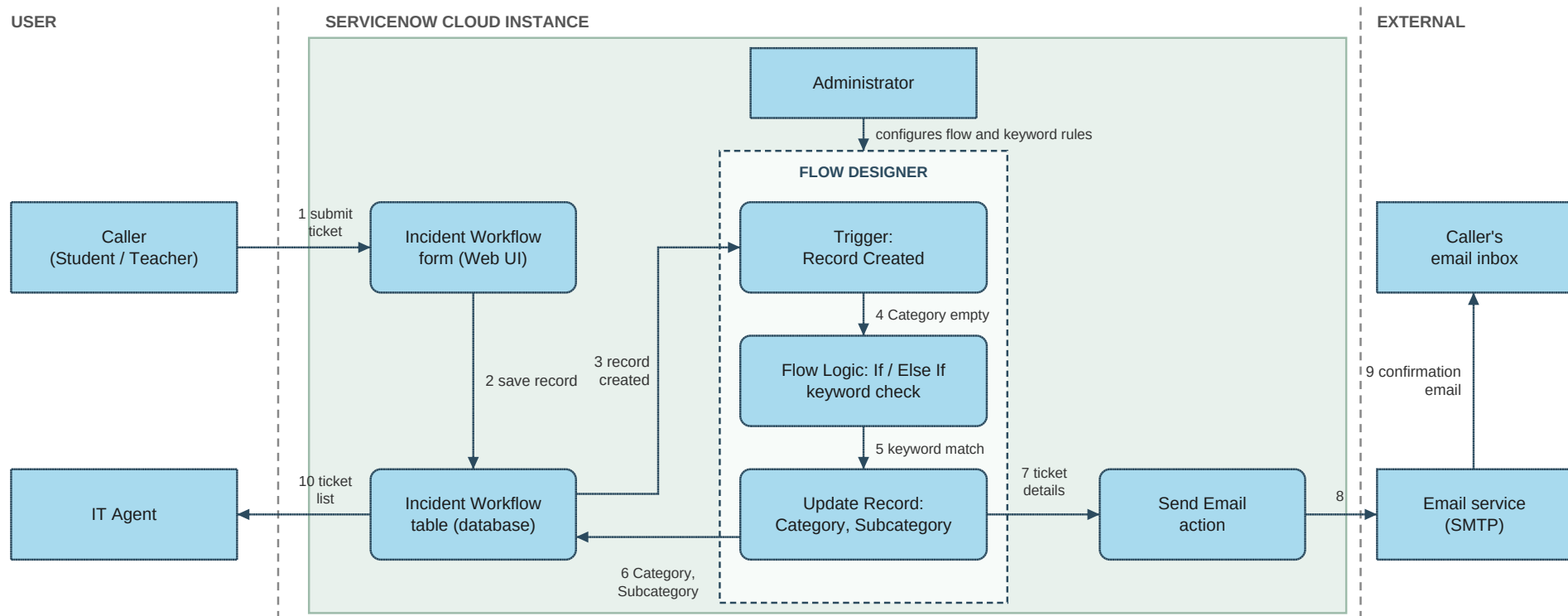


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	How the caller enters a ticket: the Incident Workflow form and list view in the browser, with only Caller and Short Description required.	ServiceNow web UI (HTML / CSS / JavaScript rendered by the platform)
2.	Application Logic-1	Trigger: the flow starts when a new record is created in Incident Workflow and Category is empty.	ServiceNow Flow Designer (Record Created trigger)
3.	Application Logic-2	Classification: If / Else If conditions check whether the Short description contains Wi-Fi / Network, Projector, Forgot password or Slow computer.	Flow Designer Flow Logic (If, Else If, 'Contains')
4.	Application Logic-3	Update Record action sets Category and Subcategory (e.g. Network / Wi-Fi, Hardware / Projector, Access / Forgot Password, Performance / Slow Computer).	Flow Designer Update Record action
5.	Database	Custom Incident Workflow table with Number (auto), Caller, Category, Subcategory, Short Description, Description, State, Assigned Group, Assigned to; dependent choices for Subcategory.	ServiceNow platform table (relational), Form Designer, Dictionary
6.	Cloud Database	The ticket data is stored in the cloud-hosted ServiceNow instance database.	ServiceNow cloud instance database
7.	File Storage	Not required: tickets hold text only and no attachments are part of the design.	Not used
8.	External API-1	Email delivery of the confirmation message to the caller's email address after the ticket is created.	Flow Designer Send Email action (SMTP email service)
9.	External API-2	No other third-party API is used.	Not used
10.	Machine Learning Model	No ML model; classification is rule-based keyword matching so it stays simple and easy to maintain.	Not used (rule-based, no-code)
11.	Infrastructure (Server / Cloud)	Application deployment on the cloud. Local Server Configuration: none, users only need a web browser. Cloud Server Configuration: ServiceNow instance, Global application scope, changes tracked in 'Project Update Set'.	Cloud: ServiceNow instance

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	No open-source frameworks are needed; the solution is built with the platform's no-code Flow Designer and form tools.	None (ServiceNow Flow Designer, no-code)
2.	Security Implementations	Users sign in to ServiceNow; role-based access controls limit who can change the table, flow and keyword rules; Caller, Assigned to and Assigned Group use reference fields to existing user and group records.	ServiceNow authentication, roles and ACLs, HTTPS
3.	Scalable Architecture	Platform-based design where new categories, subcategories and keywords are added as new choices and If / Else If branches, without code.	Flow Designer branches, choice lists, dependent fields
4.	Availability	The solution runs on the cloud-hosted ServiceNow instance, so tickets can be raised and classified at any time, not only during IT staff hours.	ServiceNow cloud-hosted instance
5.	Performance	Classification and email happen automatically when the ticket is created; simple 'Contains' conditions keep the flow light and avoid manual triage delay.	Flow Designer conditions (no scripting)